



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

Describing Meaning LI7869

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Trinity College Dublin

Semester 2 2026

What is Meaning?

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- **How do we know** what a sentence means?

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- Meaning is **surprisingly difficult** to study **scientifically**.

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- **How do we know** what a sentence means?
- Meaning is **surprisingly difficult** to study **scientifically**.
- Our goal: develop **formal tools** to understand and analyse meaning.

What does this mean?

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- **No idea** - one for the philosophers.

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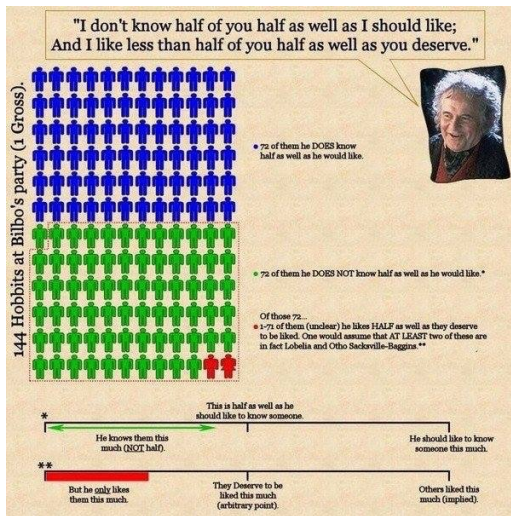
- Can we **paraphrase** it?
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- Can we **paraphrase** it?
- Why is it **difficult** to interpret quickly?
- Is it possible to describe what it means in a way that is explanatory?



What does this mean?

- *She saw the man with a telescope.*

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- Two interpretations:
 - **Instrument**: she used a telescope.
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- Same string, different meanings.

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- **Literal meaning:** a question about ability.
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- This is a case of **indirect speech acts**.

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- Something like “*A country without a language is a country without a soul.*”

Object Language vs. Metalanguage

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- Part of our job is to give explanations in a way that is **explanatory**, **useful**, or allows us to analyse or model things **scientifically** or **mathematically**.
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- Ultimately, the goal is to use tools like **set theory** and **formal logic**.

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 - **Pragmatics**

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 - sentence meaning analysed systematically,
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- Some Contrasts:
 - **Lexical semantics**
 - **Cognitive semantics**
 - **Pragmatics**
- Our focus: **truth-conditional semantics**.

Housekeeping

How the module works

- Thursdays:

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 - **16:00–18:00**

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 - Phoenix House, **Room 113**

A typical week

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- Each week we will look at:
 - an **empirical phenomenon**,
 - **formal tools** to analyse it,
 - and **cross-linguistic data** to see what varies and what is stable across languages.

Assessment

- **Problem Set 1**

Assessment

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 - Due: **March 9**

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- Worth: **40%**

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 - Due: **April 27**

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- Due: **March 9**
- Worth: **40%**

- **Problem Set 2**

- Due: **April 27**
- Worth: **60%**

Week-by-week course structure

- **Week 1:** What is Meaning?

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- **Week 12:** Modals

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- We will focus on:
 - **Entailment**
 - **Truth conditions**
 - **Tautologies, contradictions, contingencies**
 - **Validity**
 - **Limits of truth-conditional semantics**
- We will work through short examples together using a handout.

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Truth values vs. truth conditions

- A sentence can be associated with a **truth value** (TRUE or FALSE).
- A sentence also has **truth conditions**.
- **Truth value:** depends on how the world actually is.
- **Truth conditions:** what the world would have to be like for the sentence to be true.
- A sentence can have clear truth conditions even if we do not know (or cannot know) whether it is actually true.

Examples: truth conditions without knowing the truth value

- 1 *There are exactly 456 books in this building.*

Examples: truth conditions without knowing the truth value

- ① *There are exactly 456 books in this building.*
- ② *It was raining in Dublin at 14:09 on the 27th of October 1994.*

Examples: truth conditions without knowing the truth value

- ① *There are exactly 456 books in this building.*
- ② *It was raining in Dublin at 14:09 on the 27th of October 1994.*
- ③ *Every galaxy has at least one habitable planet.*

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- **Definition.** A sentence A **entails** a sentence B iff in every situation where A is true, B must also be true.

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- **Definition.** A sentence A **entails** a sentence B iff in every situation where A is true, B must also be true.
- Equivalently, A entails B iff there is no way for A to be true while B is false.
- Notation: $A \models B$.

Entailment: examples

- *John is tall and Mary is short.* $\models$ *John is tall.*

Entailment: examples

- *John is tall and Mary is short.* $\models$ *John is tall.*
- *John is tall.* $\not\models$ *Mary is short.*

Exercise A: entailment judgments

For each pair, decide whether the first sentence entails the second, the second entails the first, or neither, and briefly justify your judgment.

- ① *Mary adopted a dog.* *Mary adopted an animal.*

Exercise A: entailment judgments

For each pair, decide whether the first sentence entails the second, the second entails the first, or neither, and briefly justify your judgment.

- ① *Mary adopted a dog.* *Mary adopted an animal.*
- ② *Mary didn't a dog.* *Mary didn't adopt an animal.*

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- ③ *John is a bachelor.* *John is an unmarried man.*

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- ④ *Some students came to class.* *All students came to class.*

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- ② *Mary didn't a dog.* *Mary didn't adopt an animal.*
- ③ *John is a bachelor.* *John is an unmarried man.*
- ④ *Some students came to class.* *All students came to class.*
- ⑤ *John solved every problem.* *John solved problem 1.*

Tautologies and contradictions

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Tautologies, contradictions, contingencies

- **Tautology:** true in every possible situation.

Tautologies, contradictions, contingencies

- **Tautology:** true in every possible situation.
- **Contradiction:** false in every possible situation.

Tautologies, contradictions, contingencies

- **Tautology:** true in every possible situation.
- **Contradiction:** false in every possible situation.
- **Contingent:** true in some situations and false in others.

Exercise B: tautology, contradiction, contingent

Classify each sentence as a tautology, contradiction, or contingent.

- 1 *Either it is raining or it is not raining.*

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- ④ *Some dogs have tails.*

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- We will say an argument is **valid** if the conclusion follows from, or is provable from, the premises.
- In most cases, **validity and entailment** align, and **proof and truth** align.
- But there are important cases where these can come apart.
- The most famous results here are due to **Kurt Gödel**.

Exercise C: validity

Decide whether each argument is valid.

- 1 Premise: *John is tall and Mary is short.*
Conclusion: *John is tall.*

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- Some expressions do not seem to be the kind of thing that is simply TRUE or FALSE (or their truth depends on tricky background assumptions).

Exercise D: the limits of truth-conditional semantics

For each sentence below, discuss whether it is true, false, or whether it is not the kind of thing that is straightforwardly true or false. If you think it can be true or false, say under what conditions.

- 1 *Close the door.*

Exercise D: the limits of truth-conditional semantics

For each sentence below, discuss whether it is true, false, or whether it is not the kind of thing that is straightforwardly true or false. If you think it can be true or false, say under what conditions.

- ① *Close the door.*
- ② *Who closed the door?*

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- 2 *Who closed the door?*
- 3 *The present king of France is bald.*

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- ② *Who closed the door?*
- ③ *The present king of France is bald.*
- ④ *Sherlock Holmes lives at 221B Baker Street.*
- ⑤ *'Twas brillig, and the slithy toves did gyre and gimble in the wabe.*